Generated Question Paper

Q1. (d) State the name of the process converting ammonium ions into nitrate ions. [1 marks]

Q2. (a) The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. (i) State what is meant by the term sustainable resource. [2 marks]

Q3. (d) State the names of two products of biotechnology, other than bread, that make use of microorganisms. [2 marks]

Q4. State two advantages of using a pyramid of energy rather than a pyramid of biomass to represent a food chain. [2 marks]

Q5. (d) Suggest why reflexes occur in unconscious people. [1 marks]

Q6. (b) Fig. 6.2 shows a reproduction method used by some potato farmers. begincenter endcenter Describe the advantages of this type of reproduction in crop production. [3 marks]

[■ Diagram not found: images/potato_tubers_method.png]

Q7. (ii) State a conclusion for the results shown in Table ....
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.... [1 marks]

Q8. (c) Outline the reflex arc pathway in response to shining bright light into the eye. [3 marks]

Q9. (b) (i) State the name of the source of energy used by organism A. [1 marks]

Q10. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]

Q11. (f) Describe what happens to a fertilised egg cell before implantation in the uterus. [3 marks]

Q12. (f) Discuss the effects of non-biodegradable plastics on terrestrial ecosystems. [5 marks]

Q13. (a) Describe the possible causes of increased atmospheric carbon dioxide. [3 marks]

Q14. Table 5.1: Plants growing in Europe begincenter begintabularcc toprule Plant Number & Leaf Area (cm2) midrule 1 & 310 2 & 335 3 & 390 4 & 348 5 & 365 midrule Mean & 350 bottomrule endtabular endcenter (i) Using the data in Table 5.2, calculate the mean leaf area for plants growing in North America. Give your answer as a whole number and include the unit. [2 marks]

Q15. (a) State the name of organism A in the figure above. [1 marks]

[■ Diagram not found: images/bread_making_steps.png]

Q16. A: Blood vessel C. Blood vessel C transports blood from the to the . [2 marks]

Q17. (b) Soybean plants, Glycine max, were grown in two plots with different CO$_2$ concentrations, 370 ppm and 550 ppm. Fig. 2.1 shows average photosynthesis rates. begincenter endcenter noindent Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

[■ Diagram not found: images/photosynthesis_rate_graph.png]

Q18. (d) Scientists had higher breathing rates in the 550 ppm CO$_2$ plot. [2 marks]

Q19. (b) Complete Fig. 5.1 by stating two other organs that belong to two organ systems. begincenter endcenter noindent (c) Fig. 5.2 shows photomicrograph of part of mammalian testis. begincenter endcenter Explain why meiosis is necessary in the testes. [3 marks]

[■ Diagram not found: images/organs_two_systems.png]

[■ Diagram not found: images/mammalian_testis_micrograph.png]

Q20. (c) State the names of molecules converted into ammonium ions in the nitrogen cycle. [1 marks]

Q21. (ii) State the name of the process occurring at step 3 that causes gas bubbles to form in the dough.
[1 marks]